

Yashwanth Kasanneni

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EDUCATION

NYU Tandon School of Engineering

Master's in Computer Science

Coursework: Analysis of Algorithms, Distributed Systems, Machine Learning, Principles of Database Systems

Expected May 2027

New York, United States

WORK EXPERIENCE

Software Developer Intern

Enabled Analytics

Sept 2023 - Dec 2023

On-Site

- Developed a custom Salesforce-based order management system integrated with WhatsApp, allowing supermarket customers to browse items, place orders, and receive automated responses through a chat-based workflow.
- Built Apex, SOQL/SOSL, and Salesforce Flow automation to process WhatsApp orders in real time, validate item availability, update inventory, prevent duplicate orders, and sync order data across Salesforce records.
- Created Salesforce Lightning interfaces for supermarket staff to monitor WhatsApp orders, update fulfillment status, manage inventory, and access customer/order history from a centralized dashboard, owning the feature end-to-end from design through testing and production deployment while debugging workflow-execution issues across order placement and inventory sync.

PROJECTS

SpreadDesk - Real-Time Commodities Spread Analytics | Python, FastAPI, Redis, WebSockets, PostgreSQL, Linux

- Built a trader-facing analytics tool that ingests public EIA energy market data across natural gas, crude, refined products, and power, computing crack and spark spreads and cross-commodity relationships to surface pricing dislocations across correlated instruments.
- Engineered a low-latency Python pipeline on Linux with Redis-backed caching and WebSocket fan-out that delivers continuously updated spreads and threshold alerts to the desk, profiling hot paths to keep update-to-display latency low under bursty market data.

P2P Stock Trading Engine - Real-Time Order Matching & Settlement | Python, FastAPI, WebSockets, Kafka, Redis, PostgreSQL

- Built an event-driven peer-to-peer trading engine with market-data ingestion, WebSocket fan-out, a price-time-priority limit order book, deterministic settlement, and an append-only ledger providing auditable balances and payouts across concurrent contracts.
- Enforced idempotent order and settlement processing through unique-key constraints to prevent duplicate execution, then load-tested the in-memory matching engine under simulated burst traffic and profiled the critical path to cut p99 order-processing latency by 10x.

CourtIQ - Real-Time Streaming & Analytics Platform | Python, Kafka, Spark Structured Streaming, Airflow, Apache Iceberg, PostgreSQL, FastAPI

- Built a real-time analytics platform on a Lambda architecture, pairing a Kafka and Spark Structured Streaming speed layer for live statistics with an Airflow-orchestrated batch layer writing to Apache Iceberg for historical analysis over the same event stream.
- Served merged streaming and batch views through a FastAPI backend over PostgreSQL, ran the multi-service stack in Docker Compose, and completed a 24-point architectural gap analysis of the system's failure modes and scaling limits.

Recall - Distributed Multimodal Ingestion & Search Platform | Python, FastAPI, TypeScript, Next.js, AWS (S3, SQS, ECS), PostgreSQL, OpenSearch, Terraform

- Built a cloud-native ingestion pipeline that connects a third-party storage API over OAuth and processes files through Amazon SQS worker pools that scale on queue depth, with incremental synchronization that avoids full-account rescans.
- Designed the pipeline for reliability with idempotent job processing that safely tolerates duplicate and re-run deliveries, dead-letter queues and retry policies for failing jobs, and operational metrics across ingestion, processing, and search latency for fast failure detection.

RESEARCH AND PUBLICATIONS

[Effective Analysis of Machine and Deep Learning Methods for Diagnosing Mental Health Using Social Media Conversations](#)

IEEE Transactions on Computational Social Systems

- Analyzed 100K+ tweets and developed NLP pipelines using BERT, RoBERTa, DistilBERT, XLNet, and classical/deep-learning models for behavioral risk-pattern detection.
- Evaluated model accuracy and sentiment features to build predictive systems that identified psychological-condition signals with 97% accuracy.

SKILLS

Languages and Frameworks: Python, TypeScript/JavaScript, SQL, FastAPI, Node.js, React, Next.js, data structures and algorithms

Real-Time and Low-Latency Systems: event-driven architecture, WebSockets, SSE, limit-order-book matching, in-memory processing, performance profiling, load testing, p99 latency optimization

Streaming, Data and Reliability: Kafka, Spark Structured Streaming, Airflow, Redis, PostgreSQL, ClickHouse, OpenSearch, Apache Iceberg, idempotency, dead-letter queues, retries/replays, market-data/third-party API integration

Infrastructure, Linux and Tooling: Linux, Docker, Kubernetes, AWS (S3, SQS, ECS), GCP (Cloud Run), Terraform, Temporal, CI/CD (GitHub Actions), Datadog, AI-assisted development (Claude Code, Cursor)

Certifications: Machine Learning Specialization by Andrew Ng, Google Cloud Digital Leader